

7.6 Area of Composite Figures

Lesson Introduction

 Benchmark	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		 Learning Targets	- I can find the area of composite figures. - I can find the value of a missing dimension for a quadrilateral or a composite figure.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		Working Toward MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.	
 Essential Questions	- How do you find the area of a composite figure? - How can you use the areas of triangles and rectangles to find the area of a composite figure? - How do you use given dimensions to find the missing dimension of a composite figure?			
 Lesson Narrative	This lesson progresses from refreshing students' prior knowledge of finding the area of a quadrilateral by decomposing it to solving mathematical and real-word problems involving the area of a quadrilateral and composite figures. Students decompose composite figures to find the area. Students are then guided to use the area of a composite figure to solve for missing dimension lengths on the figure. Finally, students complete a self-evaluation to explain what they still do not understand and to ask questions about the lesson.			
 Component Summary	7.6.1 Warm-Up (~5 min.)	Students create a triangle on a coordinate plane closest to six square units (<i>SE p. 326</i>)	7.6.4 Collaboration (10 min.)	Collaboration to find missing math facts in equations and missing dimensions in rectangles and triangles (<i>SE p. 329</i>)
	7.6.2 Guided Instruction (10 min.)	Discover multiple ways to decompose a composite figure and find its area (<i>SE p. 327</i>)	7.6.5 Guided Practice (5-10 min.)	Guided practice finding missing dimension of a composite figure using the given measurements (<i>SE p. 329</i>)
	7.6.3 Guided Practice (10 min.)	Decompose composite figures into triangles and rectangles to find their area (<i>SE p. 328</i>)	7.6.6 Your Turn! (10 min.)	Individual practice finding the area of and missing dimensions in composite figures (<i>SE p. 330</i>)
	7.6.7 Wrap-Up (~5 min.)	Reflection on lesson (<i>SE p. 331</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Composite figure - Area of a composite figure <u>Additional Terms:</u> - Dimension - Composite figure - Area		- Four-Function calculator (Optional) Scratch, tracing, or graph paper (Optional) Scissors (Optional) Separate paper (Optional) Square and rectangle manipulatives	
			Additional Preparation Consider making copies of cutouts of each composite figure to allow students to manipulate the shape.	

 Teacher Tips	Common Misconceptions - Students may think a figure can only be decomposed into two shapes when more than two shapes are needed to find the area. - Students may not realize they need to use two measurements to find a third measurement that is not given. - Students with gaps in spatial awareness may struggle with decomposing a composite figure. - When more than one side does not have a measurement already labeled, students may struggle with discovering which side they need to solve for.	Things to Consider - Spend time showing students that there are different ways to divide a composite figure. Encourage students to choose ways that have the most given side lengths. - Encourage students to decompose composite figures in different ways to find the best way to calculate the missing dimension. - Consider not giving students a four-function calculator for this lesson to build student's fluency on the operations and using rational values.
	Literacy and Language Routines Think-Pair-Share In 7.6.7 Wrap-Up, students engage in a reflection to think about and then discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts related to solving for missing dimensions on composite figures.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency During 7.6.4 Collaboration, students apply their knowledge of math facts to find missing factors and area of rectangles and triangles to find missing dimensions on composite figures. Using background knowledge and discussing multiple approaches to solving for area or missing dimensions can position students to complete tasks accurately and with confidence.
	 Differentiate Instruction The intent of 7.6.2 Guided Instruction and 7.6.3 Guided Practice centers on using decomposition to find area, not finding area alone. Students who still struggle finding area should leverage On-Ramp to 6th Grade videos on finding Area. Consider having students preview this lesson by watching the On-Ramp video or have the video available to watch for students to apt-in during 7.6.6 Your Turn! questions. In the 7.6.5 Guided Practice portion of this lesson, students need to make sense of visual representations of composite figures to solve for the missing dimension. Consider providing visual and kinesthetic entry points to learning, including: - Graphing the given figures by hand on graph paper or using online tools like GeoGebra to get a better sense of how the side lengths relate to each other. - Providing graph paper for students to recreate the drawing to better understand how to decompose the shape. - Using tracing paper to trace, cut out, and manipulate composite figures; physically cutting paper to decompose the shapes and identify how to use given measurements to find missing dimensions. - Using graphic organizers or lists to keep track of steps for solving or different equations used to find area of smaller shapes within composite shape.	
Lesson Notes:		

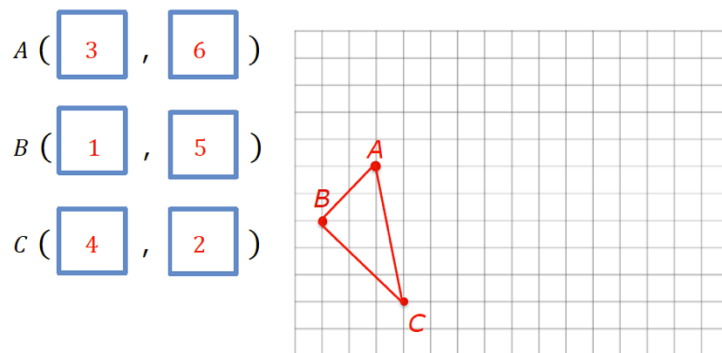
7.6.1 Warm-Up: Closest to Six

Component Overview: Students create a triangle on a graph whose area is closest to where the digits one through nine are used only one time each as coordinates. Consider using Think-Pair-Share routines on this activity, as students explore the area of a triangle and how it is related to a rectangle.



7.6.1 Warm-Up: Closest to Six

- Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.



Think-Pair-Share

Consider using Think-Pair-Share routines as students test and determine an area of a triangle closest to six square units. As students share with partners, be sure to prompt students to think about what the possible side lengths are needed to get an area of 6. When students share ideas with a class, write responses on a shared classroom whiteboard so the whole class can see ideas and make connections or identify solutions that may not work.



As a visual approach for students struggling to get started, consider having students construct a rectangle whose area is equal to 12 square units using graph or tracing paper, then cutting it in half to represent 6 square units. Then have students move the triangle around the graph until they find points that are numerically different and cover approximately the same area.

7.6.2 Guided Instruction: Area of Composite Figures

Component Overview: Students will explore how to decompose a composite figure into multiple shapes and find their areas, whose sum will be the area of the composite figure. It is important to show students how there are multiple ways a shape can be decomposed and still get the correct area of the figure.

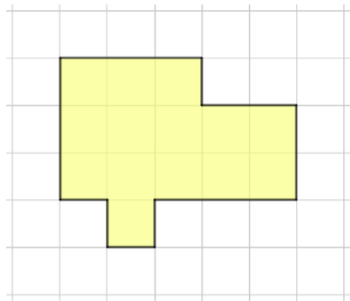


7.6.2 Guided Instruction: Area of Composite Figures



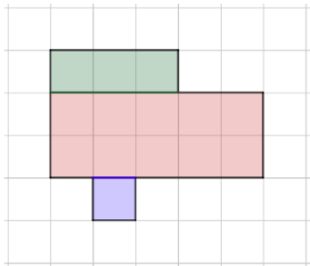
A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

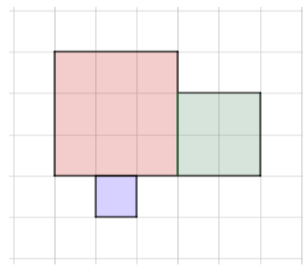


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



To highlight the multiple ways students can decompose composite figures, ask questions like:

- Is either student correct or incorrect? Why or why not?
- How are Jackson's and Danna's answers the same? How are they different?
- How would you have decomposed the composite shape?
- How could the decomposition help you find the area of the composite figure?

7.6.2 Guided Instruction: Area of Composite Figures (continued)

- Complete the table about each student's work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson	10 cm^2	1 cm^2	3 cm^2
Danna	9 cm^2	1 cm^2	4 cm^2

- To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	14 cm^2
Danna	14 cm^2

- Explain why both students got the same area of the composite shape.

Both students got the same area of the composite shape because it does not matter which way you break down a shape, the area is the same.



The **area of a composite figure** is the sum of the areas of the figure's decomposed parts.



To further scaffold question 3, ask questions like:

- Does decomposing the composite figure change the shape overall? Why or why not?
- What do you notice about the areas of the smaller shapes? What does this mean for the area of the composite figure?

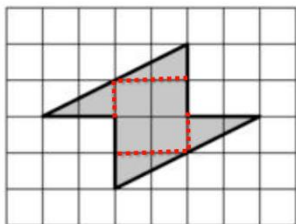
7.6.3 Guided Practice: Area of Composite Figures

Component Overview: Students will practice decomposing a composite figure. Students will be given a figure in which they can choose how to break up to find the areas of each shape in the figure to find the total area of the composite figure.



7.6.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- a. Draw lines to decompose the figure into rectangles and/or triangles.
- b. Find the area of each rectangle and/or triangle. Include units.

$1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 4 \text{ in}^2$

- c. Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.

8 in^2

- d. Find a classmate who decomposed the shape differently than you did. Describe their process.

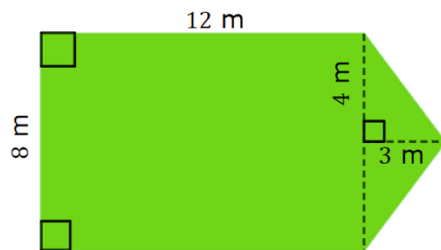
Answers will vary. My classmate divided the shape into two triangles. My classmate divided the shape into 3 triangles and 1 rectangle.



This activity provides visual entry points through the representations of the shapes shown. Auditory entry points can be provided through partner or whole-class discussion or reading aloud the questions. Consider providing cutouts of the shapes to provide kinesthetic entry points for all learners.

7.6.3 Guided Practice: Area of Composite Figures (continued)

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.



How much sod does Thomas need to buy?

$$A_{\text{rectangle}} = 8\text{ m} \times 12\text{ m} = 96\text{ m}^2$$

$$A_{\text{triangle}} = \frac{8\text{ m} \times 3\text{ m}}{2} = 12\text{ m}^2$$

$$A_{\text{total}} = A_{\text{rectangle}} + A_{\text{triangle}} = 96\text{ m}^2 + 12\text{ m}^2 = 108\text{ m}^2$$



Think-Pair-Share

Before answering question 2, engage students in a Think-Pair-Share routine. Ask guiding questions to provide further scaffolding like:

- How many shapes can you decompose this composite figure into?
- How can you use those shapes to find the area of the composite figure?

Have students first share their responses with a partner. Then share out as a class and clarify any misconceptions.



Consider prompting students to utilize a manipulative like a piece of paper for replicating this decomposition.

7.6.4 Collaboration: Finding a Missing Value

Component Overview: Students use operations reasoning to find a missing value in a multiplication or division equation, then determine how it relates to an equation using the same numbers in a different order with a different operation. Students then explore how to transfer this knowledge to finding the missing side length of a shape. Consider partnering students to engage in math discussions and check answers as they collaborate.

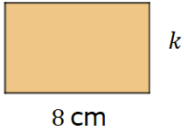
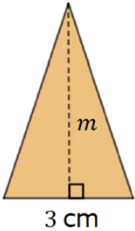


7.6.4 Collaboration: Finding a Missing Value

- Complete the table by filling in the blanks for the following math facts.

If.....	Then....
$40 \div 8 = \underline{5}$	$8 \times \underline{5} = 40$
$27 \div 3 \div \frac{1}{2} = \underline{18}$	$\frac{1}{2} \times \underline{18} \times 3 = 27$
$8 \div \frac{8}{7} = \underline{7}$	$\underline{7} \times \frac{8}{7} = 8$

- With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).

		
Area = 40 cm ²	Area = 27 cm ²	Area = 8 cm ²
$k = \underline{5}$	$m = \underline{18}$	$x = \underline{7}$



For question 1, encourage students to make connections between prior knowledge and the geometry tasks in this section by asking students:

- Which operations are inverses?
- How can inverse operations be used to find a missing number?



Use visual approaches to see the connection between the formula and diagrams shown by having students write the area formulas for each shape and substitute the given information until they are left with a variable to solve for.



For question 2, prompt student thinking about the relationship between the shapes' areas and missing dimensions by asking:

- What is the relationship between the area of a shape and the given dimension?
- How do you find the area of a triangle and rectangle?
- Where in the real world would you need to find a missing dimension?

7.6.5 Guided Practice: Finding the Value of the Missing Dimension

Component Overview: Students work individually to practice finding the missing dimension of a composite figure using given measurements. This activity combines prior knowledge of how to find a missing value in an equation with using an area equation to find the missing dimension.



7.6.5 Guided Practice: Finding the Value of the Missing Dimension

- Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of 6.875 square feet (ft^2).

- What is the area of the triangular top? Include units.

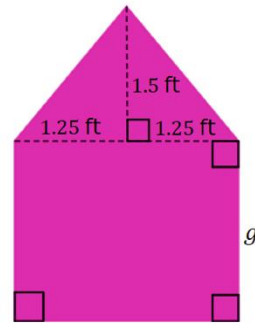
1.875 ft^2

- What is the area of the rectangular base? Include units.

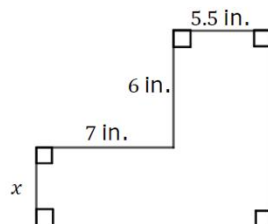
5 ft^2

- What is the height, g , of the rectangular base? Include units.

2 ft



- The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.



$$x = 4.08 \text{ in}^2$$



As students engage in question 1, provide questions to scaffold student thinking and reasoning about composite figures and finding area using properties of operations.

- How do you find the area of a triangle?
- Which side is the base of the triangle? How can you tell?
- What can you multiply by 1.25 to get the base of the triangle measurement? What's another expression you could use with addition?
- What is the formula for area of a triangle? What happens when you find $\frac{1}{2}$ of $2(1.25)$?
- How can you use this information to simplify your work before multiplying?



Students may incorrectly label sides using given measurements instead of solving for the unknown side lengths. Consider providing string as a measuring tool for students to verify measurements.



Contemplate, then Calculate

Prompt students to consider their approach to solving before calculating. Infusing this metacognitive strategy will contribute to fluency and mitigate errors.

7.6.6 Your Turn!

Component Overview: Students will demonstrate understanding of decomposing a composite figure to find the area of the shapes that make up that figure in order to find the total area using the given measurements.



7.6.6 Your Turn!

For each composite figure, find the area, in square feet (ft^2). Show your work.

1.

$$A_1 = 4 \text{ ft} \times (7.4 \text{ ft} - 5 \text{ ft}) = 9.6 \text{ ft}^2$$

$$A_2 = 5 \text{ ft} \times 2.8 \text{ ft} = 14 \text{ ft}^2$$

$$A_T = A_1 + A_2 = 9.6 \text{ ft}^2 + 14 \text{ ft}^2 = 23.6 \text{ ft}^2$$

Area = 23.6 ft^2



Consider differentiating by content with one of the following strategies:

- Choice: Allow students to choose two out of three questions to complete.
- Gridlines: Provide students with graph paper to trace each composite figure, then use the gridlines to better make sense of ways to decompose each figure.

2.

$$A_1 = \frac{(7.4 \text{ ft}) \times (2 \text{ ft})}{2} = 7.4 \text{ ft}^2$$

$$A_2 = 5.4 \text{ ft} \times 2.8 \text{ ft} = 15.12 \text{ ft}^2$$

$$A_T = 7.4 \text{ ft}^2 + 15.12 \text{ ft}^2 = 22.52 \text{ ft}^2$$

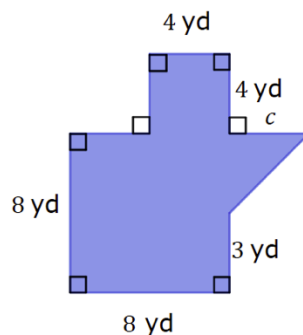
Area = 22.52 ft^2



Challenge students who finish early to create their own composite figure, including dimensions with at least one missing dimension, to trade with a partner and submit for extra credit.

7.6.6 Your Turn! (continued)

3. The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.



Not drawn to scale

$$c = 3 \text{ yd}$$



As a kinesthetic entry point, consider giving students a sheet of paper to draw the shapes that compose the composite figure. Have students cut out the shapes to visualize which measurement is missing. Also consider having students graph the given figures by hand or on GeoGebra to get a better sense of how the side lengths relate to each other.



For individual students who may have difficulty getting started, initiate geometric reasoning and problem-solving by asking:

- *How did you approach decomposing this figure? What were your reasons for decomposing it that way?*
- *How can you tell what the height of the triangle is? If you know the height and you are solving for the missing dimension c as the base, what other value do we need to solve for c ?*

7.6.7 Wrap-Up: Reflection

Component Overview: This Wrap-Up positions students to reflect on the lesson's learning targets and provide an opportunity to ask remaining questions about finding area of composite figures and/or finding missing dimensions. Consider using an auditory, collaborative approach to this reflection by having students share their questions with a partner and seek answers to clarify any misconceptions.



7.6.7 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about ...
Answers will vary.
2. One question I still have about today's lesson is ...
Answers will vary.



Think-Pair-Share

After students have had a few minutes to consider their responses, allow students to turn and talk to discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts for themselves or others.